09 The Adaptation Room

Multi-trait selection vs. drift

CORE PATH · BSC	EXTENSION · MSC	BEFORE THIS ROOM
about 35 min	**+ 30 min**	The Drift (04), Selection (07) and Mutation (08) rooms. The Branching Room (02) explains the tanglegram.
steps 1–4	steps 5–7	

OPEN THE ROOM

`https://evolinus.github.io/zooe/teaching/evolutionary-laboratories/#adaptation`

Three related founder fish each split into a Neutral lineage, where nothing is selected, and a Habitat lineage living in a Stream, Pond or River. Five traits mutate and are sorted by drift and selection over thousands of generations. Because every Habitat lineage has a Neutral twin from the same ancestor, you can see exactly what selection adds.

LEARNING OBJECTIVES

After the core path you should be able to

- Use a Neutral control to measure what selection contributes beyond drift.
- Explain convergent evolution: unrelated lineages becoming alike in the same habitat.
- Show that mutations arise at the same rate whatever the habitat; selection changes which ones fix.
- Explain why convergence can mislead a tree built from appearance.

The extension adds

- Use the waiting times $1/\mu$ and $1/(\mu s)$ to predict the pace of adaptation.
- Explain, using $4Ns$, why small populations fix more changes but sort them worse.
- Design and interpret a controlled comparison using the room's presets.

SUGGESTED ROUTES

- **45-minute practical (BSc):** steps 1–4 and questions 1–3. Runs take a few seconds each.
- **Revision session for Part III:** this room after Selection and Mutation, using step 3 as the control-experiment discussion.
- **MSc:** steps 5–7 and questions 4–6 (about 30 minutes).

How to work through it. Every step asks you to *predict* before you press anything, and to *explain* afterwards. Write the prediction down and commit to it: a wrong prediction you can account for teaches more than a right one you guessed. Runs are random, so where a step says so, run it more than once before you decide what happened.

1 Neutral against Habitat

12 min

SET UP

The defaults: G = 5000; Gigi in the Stream, Mario in the Pond, Nani in the River; every s = 0.20, every μ = 0.010, every N = 50.

PREDICT FIRST

For each founder, which lineage will have travelled further from the ancestor by the end: the Neutral one or the Habitat one? By a little, or by a lot?

RUN

Press **Run Simulation** and watch the two lineages of each founder evolve side by side.

LOOK AT

the table *Each lineage against its own founder*: a row for Neutral, one for Habitat, and the mean of each row.

EXPLAIN

Both lineages had the same mutation rate and the same population size. What made the difference?

2 Same habitat, different founders

10 min

SET UP

Press the preset **Same habitat, different founders** (all three Habitat lineages in the Stream). It runs by itself.

PREDICT FIRST

The three founders started different. Will their three Habitat lineages end up more alike than their three Neutral lineages, or less?

RUN

Let the preset run, and read the status line.

LOOK AT

the *Final divergence matrix*, comparing the three Habitat fish with one another and the three Neutral fish with one another.

EXPLAIN

Name the pattern you see. Why does a shared habitat make unrelated lineages converge?

3 The control experiment

8 min

SET UP

Press the preset **Turn selection off** (every $s = 0$).

PREDICT FIRST

Each Habitat lineage is now a second Neutral lineage. How big will the gap between the two rows of the table be?

RUN

Let it run. Then run it once more with **Run Simulation**.

EXPLAIN

Why is this the right control for step 1? What does any gap that remains tell you about a single run?

4 Does the tree see through convergence?

5 min

SET UP

Run the **Same habitat, different founders** preset again.

PREDICT FIRST

Will the reconstructed tree group the Habitat lineages by ancestry (each with its own Neutral twin) or by habitat (the three Stream fish together)?

RUN

Scroll to the *Tanglegram Comparison*.

EXPLAIN

Why is a tree built from final appearance at risk when unrelated lineages share an environment?

EXTENSION · MSC

5 Small against large populations

12 min

SET UP

Press the preset **Small vs large population**: three identical founders, all in the Stream, $N = 10$, 50 and 200, $s = 0.05$.

PREDICT FIRST

Which population will fix the most changes? Which will end up best adapted?

RUN

Let the preset run, then run it twice more.

LOOK AT

the *Sel.* – row of the trait tables and the *Habitat adapted* row of the table, not the Habitat Δ row.

EXPLAIN

Use the ratio of selected to neutral fixation, $4Ns$ (2 at $N = 10$, 40 at $N = 200$), to explain why the small population travels as far but less of its journey goes the way the habitat prefers.

6 Supply against fate

10 min

SET UP

Restore defaults, then run once.

PREDICT FIRST

With $\mu = 0.010$ per trait per generation, how long should a locus wait for a new mutation? About half of mutations step the way the habitat prefers and fix with $u \approx 2s$; how long should it wait for an *adaptive* substitution? Will Habitat lineages receive more mutations than Neutral ones?

RUN

Open the trait tables and compare the counts of new mutations in Neutral and Habitat lineages.

EXPLAIN

Mutations arrive at the same rate in both. Where, then, does the difference between the lineages come from?

7 Starve it of mutations

8 min

SET UP

Open *More parameters* and set every μ to **0.002**, keeping $G = 5000$.

PREDICT FIRST

Using $1/(\mu s)$, how long is the wait for one adaptive substitution per trait now? Will the Habitat lineages reach their habitat's optimum by generation 5000?

RUN

Run twice.

EXPLAIN

When is adaptation limited by the supply of mutations rather than by selection? How would standing variation (present before the run started) change that?

DISCUSSION AND WORKSHEET QUESTIONS

1. BSc Dolphins and ichthyosaurs share a streamlined body. Explain how this room models that, and why their shape says little about their ancestry.
2. BSc Why does the room need a Neutral lineage at all? What would you be unable to conclude without it?
3. BSc "The stream causes slender-bodied mutations." Correct this statement using what you saw in step 6.
4. MSc Real traits are polygenic, with many loci of small effect. How would that change the speed and repeatability of adaptation in this room?
5. MSc A small island population shows unusual traits. How would you test whether they are adaptations or the products of drift?
6. MSc Explain why a larger population "buys no extra mutations" in this room. Under what definition of mutation rate would a larger population get more?

CHECKING YOUR REASONING

Read this after you have written your own answers. These are what the model should show on most runs — not the only acceptable explanations.

- **Step 1.** The Habitat lineages travel much further from their founders than the Neutral ones. The gap is what selection adds on top of drift.
- **Step 2.** The three Habitat fish end up much more alike than the three Neutral fish: convergent evolution, produced by shared selection rather than shared ancestry.
- **Step 3.** The two rows come out similar; any remaining gap is the size of one run's chance difference, which is why a control is needed.
- **Step 4.** The reconstruction tends to pull the Habitat lineages together, misplacing them away from their own Neutral twins.
- **Step 5.** The small population fixes many changes, many of them in directions the habitat does not prefer; the large one adapts most faithfully. Measured over many runs about 1.9, 3.3 and 3.5 of 4 traits end adapted at N = 10, 50 and 200.
- **Step 6.** A new mutation every $1/\mu$ = 100 generations per locus; an adaptive substitution about every $1/(\mu s)$ = 500. Neutral and Habitat lineages receive mutations at the same rate; what differs is which ones fix.
- **Step 7.** At μ = 0.002 the wait for an adaptive substitution is about 2,500 generations per trait, so by 5000 the lineages are only partly adapted: adaptation becomes limited by supply.